J Math Teacher Educ (2014) 17:369–392
DOI 10.1007/s10857-013-9265-0

Do prospective mathematics teachers teach who they say they are?

Sonja van Putten · Gerrit Stols · Sarah Howie

Published online: 28 January 2014
© Springer Science+Business Media Dordrecht 2014

Abstract In this case study, the professional mathematics teacher identity (PMTI) of final year mathematics education students is investigated in terms of their self-perceived and actualised identity. These prospective teachers were required to discuss and describe their own PMTI in terms of three aspects: mathematics specialisation, teaching-and-learning specialisation, and caring. Subsequently, they were observed in the classroom, where the actualisation of their PMTI was considered in terms of the same three. The participants' perceptions of their own PMTI and the actualisation of that PMTI in the classroom were found not to be congruent. While their self-perceptions regarding their prowess as Mathematics Specialists were accurate, since this is concretely tested as part of their studies, their self-perceptions as teaching-and-learning specialists and particularly as Carers, were not verifiable in their classroom practice. Espoused theory, theory that the individual perceives as true and valid, and which may thus be seen as intrinsic to their PMTI's, is not necessarily enacted.

Keywords Professional mathematics teacher identity · Classroom actualisation · Theory-in-use

Introduction

The professional identity of the person who teaches is an essential factor in determining the success of what happens in the classroom, if we are to believe the dictum that says we teach who we are. Mathematics education in South Africa, despite the many changes in education since 1994, remains in crisis. While effort and money have been expended on improving facilities and teachers' knowledge of mathematics, the crisis persists. Arends and Phurutse (2009) believe that a difference can be made to the state of mathematics

S. van Putten (✉) · G. Stols · S. Howie
Department of Science, Mathematics and Technology Education, Faculty of Education, University of Pretoria, Groenkloof Campus, Leyds Street, Pretoria 0181, South Africa
e-mail: sonja.vanputten@up.ac.za

education by "good teaching": "The study of teachers and teaching deserves much more attention than it has been given, particularly in the light of growing empirical evidence that good teaching makes a huge difference to learning regardless of the socio-economic status of the learners" (p. 45). Good teaching focuses on the teacher: who is this person? This research purports to gain insight into the professional identity of the *prospective* mathematics teacher—investigating the phenomenon at its origin—in terms of how the individual perceives her professional identity and how that identity is actualised in the classroom.

Literature review

There are many definitions of professional teacher identity (PTI), so many that the concept has been described as amorphous; it is complex and multifaceted (Day 2002; Chevrier et al. 2007; Vloet and van Swet 2010) and composing a definition for it is a problematic exercise (Beauchamp and Thomas 2009). For example, Gee (2000) said, "Being recognised as a certain "kind of person", in a given context, is what I mean here by "identity"" (p. 99). He links "who" to "where". Van den Berg (2002) situates professional identity in the interaction between personal experiences and the education environment; Walkington (2005) links professional identity to beliefs—beliefs one has about teaching and being a teacher; beliefs that are continuously formed and reformed through experience; Shapiro (2010) believes professional identity is based on affect—we feel that we have chosen this field and it has chosen us. Beijaard (2010, personal communication) speaks of professional identity as an interaction between the personal and the professional. PTI is complex in that it is made up of personal as well as social aspects that come together in a construct that encompasses knowledge and beliefs, emotions and relationships, and contexts and experiences.

This being said, the teacher should be recognised as a complex persona who is affected by societal and personal interactions that result in the "self-hood" (Cardelle-Elawar et al. 2010) that she brings into the classroom. Teacher's identity, according to Varghese et al. (2005), is a "profoundly individual and psychological matter because it concerns the self-image and other-image of particular teachers" (p. 39). The context (the social aspect) in which the individual reasons, makes decisions, acts and operationalises her PTI (the psychological aspect), plays a vital role in the development of PTI. The fact that PTI is responsive to social context implies participation in communities of practice (Lave and Wenger 1991; Wenger 2000). These communities cohere because of three defining properties: joint enterprise (they are doing something, e.g. learning about teaching mathematics), mutual engagement (they are working together, e.g. as a group of fourth year students), and a shared repertoire (shared resources that may be social, physical, historical and so on, for example, they may attend the same classes taught by the same lecturers). Participation in such a community of practice is identity-linked. In fact, Wenger sees identity as the "who we are" that is continually being developed in our own minds and in the minds of those with whom we interact in such a community. PTI is developed in a particularly *specialised* society: schools and teacher training facilities. For the purpose of this study, PTI is seen as the crossroads between the personal and the social self, the "who I am at this moment".

PTI is an important tool for studying education issues (Gee 2000), since "we teach who we are". The term "Mathematics Teacher Identity", described by Bohl and van Zoest (2002) as a unit of analysis, can also be applicable to those who, although they teach mathematics from time to time or for a period, are in fact not professional mathematics

teachers—they may have been co-opted into teaching the subject because there is no one else to do so in a particular school, or some such circumstance. In South Africa, this happens frequently. Graven (2004), for example, tells the story of some teachers that she worked with:

> For example, Moses explained that it was not considered politically acceptable as a black student to study mathematics when he was at school and college. Rather, one had to study history and other subjects considered important for the struggle against apartheid … Moses had therefore studied to become a history teacher but became a teacher of mathematics due to the shortage of mathematics teachers. Another teacher, Barry, despite having taught mathematics and headed a mathematics department for many years, explained that he was not a mathematics teacher since he did not 'even' study mathematics at high school. He called himself an art teacher since this is what he had studied … Similarly … Beatrice used to introduce herself as 'the music teacher' despite teaching predominantly mathematics classes. These examples illustrate an effect of South Africa's apartheid history. (p. 189)

The term professional mathematics teacher identity (PMTI) is posited in this research as involving an individual who has studied the subject for the specific purpose of teaching it. Mastery in this profession, says Graven (2004), "involves becoming confident in relation to … [inter alia] one's identity as a professional mathematics teacher" (p. 185). PTI is subsumed in PMTI, and therefore shares its characteristics. Hodgen and Askew (2007) speak of a "strong disciplinary bond" (p. 484) which a teacher has with the subject she teaches. It is this bond with mathematics that is central to PMTI, a bond that includes a view of the subject and beliefs regarding the subject, and even emotions related to the subject. Leatham and Hill (2010) call this "mathematical identity" (p. 226) which they define as "an individual's relationship with mathematics". Therefore, if PTI can be defined as "who I am at this moment", i.e. the self that is the sum total of past and present experiences both personal and social, then PMTI is "who I am at this moment in this context" in which PTI is situated within that "relationship with mathematics" which includes beliefs and emotions associated with the subject, as well as "the ways they [mathematics teachers] see themselves as subject matter experts, pedagogical experts, and didactical experts" (Beijaard et al. 2000, p. 751).

PTI then is not a singular, unitary phenomenon, but has multiple constituents that help to make it unique for each individual (Volkmann and Anderson 1998; Coldron and Smith 1999). We believe that, rather than different *identities* coming to the fore in different contexts as Stronach et al. (2002) suggest, it is more accurate to say that different *aspects* of identity are more prominent in certain contexts. Beijaard et al. (2004) refer to these aspects as sub-identities that arise through the variety of contexts and relationships in which a teacher might live and work. While academics are not in agreement regarding appellation of this phenomenon, they are unanimous about the subdividedness of PTI.

Theoretical framework

Beijaard et al. (2000) identify three specific aspects of the mathematics teacher that may be seen as sub-identities or clusters within her PMTI, since the "who I am at this moment" is different in each one of these aspects. Beijaard et al. (2000) in their study of PTI were "inspired" (p. 751) by the work of Bromme (1991), from which they drew the idea that teachers derive their professional identity from "the ways they see themselves as subject matter experts, pedagogical experts, and didactical experts" (p. 751). Beijaard et al.

explain that, in Europe (they were based in Holland), these concepts are "relevant components of models and theories of teaching on the basis of which (prospective) teachers organise their work".

For this study, this model posited by Beijaard et al. (2000) was used: that teachers derive their professional identity from their perceptions of themselves as specialists regarding the subject matter (*subject* content knowledge and skills); teaching-and-learning (the knowledge and skills related to the preparation, execution, and evaluation of the *teaching-and-learning process*) and caring (the knowledge and skills required to undergird and *support the socio-emotional and moral* development of learners [the choice of the term "carer" rather than "nurturer" was the result of discussion with Beijaard (2010, personal communication)]. While Korthagen (2004), provides a theoretical model for investigating "what is a good teacher", describing his model as an onion and calling it "levels of change" (p. 87), and Van Zoest and Bohl (2005) offer a framework featuring "Aspects of Self-in-Mind" and "Aspects of Self-in-Community" (p. 332), we chose Beijaard et al.'s model for its facilitation of study of specific aspects of teacher identity, and because it resonates with national education policy in South Africa. According to the National Policy Framework for Teacher Education and Development in South Africa (DoE 2006), the following aspects are what should be seen in a good teacher. She must reveal herself to be

- a specialist in a particular learning area, subject or phase;
- a scholar and lifelong learner; and
- a curriculum developer;
- a specialist in teaching and learning;
- a specialist in assessment;
- a leader, administrator, and manager;
- a professional who plays a community, citizenship, and pastoral role.

The first three aspects are linked to subject specialisation and being an expert in terms of what is required in terms of a particular subject; the next two deal with expertise in the skills necessary for teaching and learning; and the last two refer to the importance of the caring, guiding, and leading aspects of being a teacher. The National Council of Teachers of Mathematics (NCTM 2008) says much the same thing: a good mathematics teacher must know mathematics well and must have the skills and strategies to guide learners' understanding and learning.

In order to further elucidate the study of PMTI in practice, i.e. in the classroom, aspects of models developed by Ernest (1988) and Thompson (2009) were also included in the theoretical framework. Ernest (1988) and Thompson (2009) found a distinct link between the individual's beliefs regarding the subject as well as her understanding of mathematics, and the way that person teaches. Ernest describes three instruction modes used by mathematics teachers: Instructor: Skills mastery with correct performance; Explainer: Conceptual understanding with unified knowledge, and Facilitator: Confident problem posing and solving. Using Thompson's model, teaching-and-learning was analysed in terms of the teacher's recognition of evidence of understanding; teacher/learner-centeredness of classroom practice, flexibility/rigidity in adherence to lesson plan. Visually, this framework can be represented as follows (Fig. 1):

Each of the three parts of PMTI is linked to at least one observable aspect. These links are described in the following three sections.

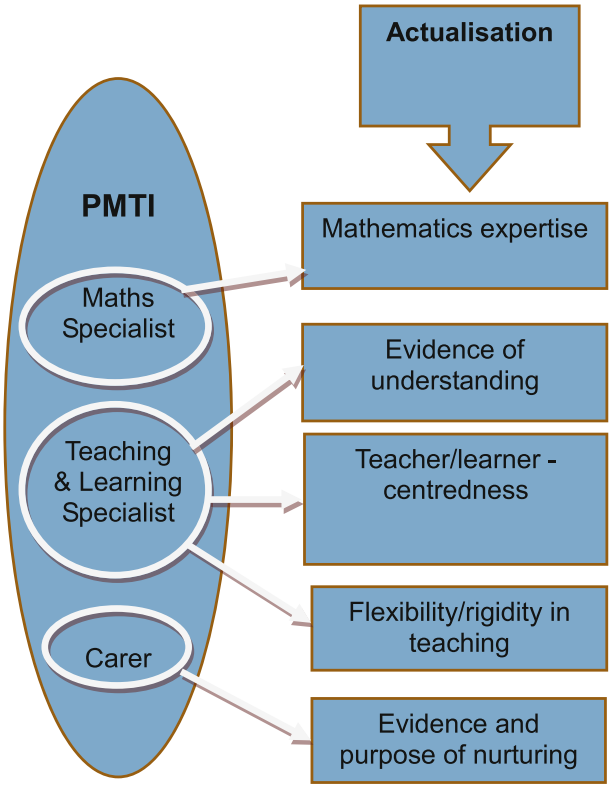

Fig. 1 Conceptual framework for PMTI and its actualisation

Mathematics specialist

The literature indicates that there is a strong correlation between the teacher's knowledge of mathematics and successful classroom practice (Hill et al. 2005; Ball et al. 2008; Wilkins 2008; Pang 2009). Therefore, subject expertise is an important aspect of the PMTI of a good mathematics teacher.

In this study, mathematics expertise is investigated as the prospective teacher's ability to deal with the actual mathematics in the classroom—both in terms of what she was teaching, and in terms of her answers to questions that arise. As part of successful classroom practice, the importance of the teacher's responses to learner questions is agreed upon by academics (Chin 2006; Ainley and Luntley 2007; Darling-Hammond and Richardson 2009). Specifically, the accuracy and comfort with which she handles the mathematical concepts is studied. This facility with the subject is called "mathematical knowledge for teaching", by Hill et al. (2005):

> [W]e mean the mathematical knowledge used to carry out the *work of teaching mathematics*. Examples of this "work of teaching" include explaining terms and concepts to students, interpreting students' statements and solutions, judging and correcting textbook treatments of particular topics, using representations accurately in the classroom, and providing students with examples of mathematical concepts, algorithms, or proofs. (p. 373)

Teaching-and-learning specialisation

Here, the skills that make a teacher able to teach effectively are in question: "sophisticated teaching" is required by society's demands for complex and analytical skills (Darling-Hammond and Richardson 2009).

With regard to the actualisation of the individual's conceptions of teaching-and-learning, Thompson (2009) examined three observable aspects:

1. the locus of control, which she saw as where the control of the activities in the classroom lay, seen in this study as learner/teacher-centeredness;
2. what is recognised as evidence of understanding. This is a significant aspect of effective classroom practice (Fisher and Frey 2007). One of the skills a good mathematics teacher should have is the ability to determine whether learners have understood while the lesson is in progress, so that immediate measures can be taken where misunderstanding is evident;
3. how flexible the participant was in her teaching. This flexibility she related to planning. While planning is an important task (So and Watkins 2005), a good mathematics teacher should be able to change her instruction based on the way the lesson unfolds as learners raise concerns and put forward ideas (NCTM 2008). This adaptability signifies the ability to notice and interpret what is happening in the classroom in order to promote learning (van Es and Sherin 2008).

Caring

In considering this aspect of PMTI, the individual's perception of herself as Carer in terms of her interaction with learners is investigated. Caring is associated with the affective aspects of classroom practice. Authors such as Zembylas (2003), Flores and Day (2006) and Shapiro (2010) recognise that this aspect is not only part of PTI, but is significant in effective teaching. Caring lies at the base of the relationship the teacher has with the learners and vice versa, which leads to the support of learners in their non-cognitive development. Kunter and Baumert (2006) refer to such caring as "personal learning support" (p. 235) and describe it as a quality dimension in teaching.

This "quality dimension" can be observed in the way the teacher relates to the learners as individuals and what the nature of their interaction is. In her dealings with the learners, the teacher's belief regarding the purpose of caring is made manifest through her actions—does she relate to the learner on an academic level in order to promote learning, or is her concern with "the socio-emotional and moral development of learners" her motivation? Answering these questions provides a description of the teacher as carer.

Thus, PMTI is investigated in terms of three aspects—Mathematics Specialist, Teaching-and-Learning Specialist, and Carer. The actualisation of these aspects is analysed through both Ernest's and Thompson's categorisations.

Problem statement

We teach who we are (Hamachek 1999; Palmer 2007). Thus, who I am as a teacher can be seen in my teaching. However, in this study, the link between who I am, and who I think and say I am, is under scrutiny. The following question is posed: Is the self-perceived PMTI, the self that is actualised in the classroom? According to Argyris and Schön (1974),

> When someone is asked how he would behave under certain circumstances, the answer he usually gives is his espoused theory of action for that situation. This is the theory of action to which he gives allegiance, and which, upon request, he communicates to others. However, the theory that actually governs his actions is this theory-in-use (p. 6).

Therefore, it is possible that espoused theory, which is intrinsic to PMTI, may not be the same as the theory-in-use, which is intrinsic to the PMTI as actualised in the classroom. On the one hand, Beijaard et al. (2000) state that the participant is the person best able to describe their PMTI; on the other hand, Palmer (2007) says we teach who we are. It is possible that a discrepancy exists between these two: that the personal perception of PMTI is not the same as the actualised PMTI.

Research design and methodology

This is an explanatory, interpretive case study in which non-observational data was provided by a biographical questionnaire and interviews with the participants for the purpose of investigating the individual's own perceptions of her PMTI. This study aims to investigate the congruence between "Who we say we are" and "Who we are when we teach". Through observation of PMTI-in-action in the classroom, the "picture" is completed by allowing comparison between the individual's perceptions of their PMTI and the practical outworking thereof. The motivation for this research design is that while the prospective teachers can express their ideas about how they teach and who they are as mathematics teachers in the questionnaire and interviews, the classroom observations were designed to give insight into what Maxwell (1996) calls "theory-in-use" (p. 76). He found that participants' perspectives are sometimes not shared openly in interviews, but that such perspectives become clear when watching the participant in action in the classroom.

Context and sample

At the University of Pretoria, the BEd (Bachelor of Education), a 4-year degree, is structured so that the elective subjects, such as mathematics, are taken alongside of education modules and other professional studies such as educational psychology across the first 3 years of study. During their fourth year, the prospective teachers spend the second and third quarters at schools doing their "internship" or teaching practicum. The prospective teachers who are preparing to teach in the further education and training (FET) phase (Grades 10–12) are trained to be Mathematics Specialists and are therefore those who, in theory, are not only *able* but who also *desire* to teach mathematics to learners who have chosen to continue with the subject to Grade 12 level. It is this group of prospective teachers who form the sample for this study. Purposive maximum variation sampling was implemented in selecting the six participants for this study: three female (White, Black, and Indian) and three male (one White and two Black).

Data collection: strategies and instruments

The prospective teachers were initially asked to complete a questionnaire which was a translated and adapted version of a questionnaire created by Prof. Douwe Beijaard in his investigation of "experienced secondary school teachers' current and prior perceptions of

their professional identity" (Beijaard et al. 2000, p. 749). The first section of the questionnaire supplied biographical data such as sex, race (by virtue of surnames), and type of high school—rural/city, formerly "Model C" (previously exclusively "white" schools)/ previously disadvantaged school (formerly black schools). This allowed a spreadsheet to be created, giving insight into the constitution of the sample and providing the basis for the selection of the subsample. The second purpose of the questionnaire and one for which the questionnaire was designed by Beijaard et al., was to "explore the way teachers see (and saw) themselves as subject matter specialists, didactical specialists, and pedagogical specialists" (ibid., p. 749), and thus provided a base for discussion in the initial interviews, as well as a basis of comparison between the participants' PMTI beliefs and their practical outworking in the classroom. This purpose was achieved through a ranking exercise indicating their perceptions of the importance of Subject Specialist, Teaching-and-learning Specialist, and Carer within their own PMTI. They were then required to explain their ranking.

Prior to the commencement of the practicum, individual interviews were conducted with each of the sub-sample members. These interviews were semi-structured, and the questions were designed to further clarify and provide depth and insight into the beliefs expressed and explanations are given in ranking exercise of Subject Specialist, Teaching-and-learning Specialist, and Carer the questionnaire. The participants were again interviewed individually after the completion of the two-term teaching practicum. The semi-structured interviews held at this point therefore yielded data regarding the overall practicum experience, as well as insights into tendencies and behaviours observed in the videoed lessons. The interview questions were designed to provide information about the participant's perceptions of themselves in relation to the subject mathematics, how it should be taught and learnt and the extent of their involvement as Carers of their learners. Two lessons taught by the prospective teachers were observed and digitally recorded. The lessons were part of the ordinary teaching day of each of the participants, and the video camera was stationed at the back of the classroom on each occasion, so as to be as unobtrusive as possible.

The recorded and transcribed interviews and class observations were member-checked to ensure that no misrepresentation took place. Qualitative research by its very definition implies *human* involvement in a very personal way. The researcher is *there*; "Interviewer neutrality is a chimera" (Cohen et al. 2000, p. 121). The researcher records and observes. Nevertheless, every effort was made to analyse and interpret the data in an unbiased way.

Data analysis

The biographical data in the questionnaire was tabulated in Excel and used in the sample selection process. Data provided in the ranking exercise where the participants indicated the relative importance given in their PMTI to Subject Specialisation, Teaching-and-learning Specialisation, and Caring was tabled alongside of the biographical data in Excel and used as a point of discussion during the initial interviews. The written responses in the questionnaire were copied into Atlas.ti and coded, using Open Coding. Twenty codes were generated inductively, bearing in mind the exigencies of the research questions and the conceptual framework.

Recordings of the interview data were professionally transcribed, without grammatical corrections or exclusion of ums and other verbal eccentricities. The transcriptions were then coded in Atlas.ti, using data-driven coding derived from a thematic content analysis.

The content was analysed with the conceptual framework in mind. The coding remained open, but began with some ideas of what to look for. Gibbs (2007) explains as follows:

> If your project has been defined in the context of a clear theoretical framework, then it is likely that you will have some good ideas about what potential codes you will need. That is not to say that they will be preserved intact throughout the project, but at least it gives you a starting point for the kinds of phenomena you want to look for when reading the text. The trick here is not to become too tied to the initial codes you construct. (p. 46)

In the analysis of the data, in an effort to follow Gibbs' advice, the codes were used as a guide for searching the text. For the initial interviews (prior to the teaching practice), 57 codes were generated, and for the second interview set, 74. The number of codes created was a function of the desire to code even nuances of meaning. The second set of individual interviews with the participants generated an extended set of codes because these interviews were conducted at the end of the long teaching practicum, and extensive questioning could take place regarding the participant's view of the subject and their experiences in the field. These codes also include those that were deduced from discussion of the video clips.

The videos were also imported into Atlas.ti. Both deductive and inductive coding was used initially as Open Coding, and then as Code by List: the elements in the conceptual framework were used as broad code subjects, such as "Evidence of Understanding"—hence the deductive aspect of the coding; then a variety of sub-codes were created, drawn from what was said—hence the inductive aspect. Twenty-three codes were initially generated. These codes were then used to tag scenes in the videos in order to organise and facilitate discussion of the videos with the participants. Then, during the individual interviews that were held subsequent to the completion of the practicum, the video clips of themselves teaching were shown to each candidate and together the prospective teacher and interviewer discussed the videos and codes were associated/generated in terms of the actualisation descriptors in the conceptual framework. This process led to the generation of the 74 codes mentioned above. These codes are presented in Table 1; the wording of certain codes has been extended for clarity of understanding.

Participants were invited to scrutinise the data collected, and two experienced researchers were asked to corroborate the interpretation of the analysis. Every effort was made not to filter the meanings and perspectives of the participants through the framework of our own perspectives, but rather to allow the participants to reveal and justify their viewpoints without the restriction of what Maxwell calls "leading, closed, and short answer questions" (1996, p. 90). Creswell and Miller (2000) define validity in this sort of context as: "how accurately the account represents participants' realities of the social phenomena and is credible to *them*" (p. 124) (emphasis added).

Results

The six prospective teachers are presented below in terms firstly of the insights gained through the interviews and then though observing them in the classroom. Thus, self-perceptions are presented alongside of actualised PMTI in classroom practice. Each presentation of a participant ends with a brief discussion of the data.

Table 1 Inclusion criteria for coding individual prospective teacher interviews and video analysis

Inclusion criteria	Codes
Beliefs about the subject mathematics	Attitude towards maths: passion
	Attitude: enjoyment of challenges
	Reason for attitude: understanding, wanting to share
	Reason for attitude: belief in creativity
	Purpose of maths: learning to think
	Purpose of maths: mental development
	Purpose of maths: real life practicalities
	View of mathematics: uncertain, theoretically dynamic
	View of maths: potential for creativity
	View of maths: procedural
	View of maths: real-world use
	View of maths: way of thinking
	View of maths: problem solving, reasoning
	View of maths: source of belief—university training
Actualisation of PMTI	Perception of self: dramatic
	Perception of self: practical
	Own description of style
	Goal: lesson should be fun
	Goal: marks + understanding
	Goal: reasoning skills and understanding
	Overall perception of video
	Weight of beliefs: content versus caring
	Weight of beliefs: independence of motivation—what drives me to …
	Weight of beliefs: understanding versus need to finish
	Weighting: university training versus school experience
Evidence of understanding	Choir response
	Visual; questions; books
	Asking questions
	Make learners explain
Flexibility/rigidity in teaching	Flexibility: answering learners
	Flexibility: depart from plan
	Planning: gives flexibility
	Planning: keep learners busy all the time
	Planning: staying ahead of sharp kids
	Planning: structure
	Planning: time management
	Planning: to get content right

Table 1 continued

Inclusion criteria	Codes
Expertise in mathematics/teaching-and-learning/caring	Need to be right
	Creativity: linking to real world
	Creativity: using manipulative
	Creativity: only possible in certain topics
	Creativity: reason—interest and functionality
	Certainty: content
	Certainties: knowledge, care
	Certainty: atmosphere of comfort
	Certainty: teaching for understanding
	Uncertainty: content
	Uncertainty: learners understanding
	Uncertainty: content not learnt as learner
	Uncertainty: running out of ways to explain
	Uncertainty: technical
	Locus of control: discipline
	Didactics: challenges
	Didactics: different methods
	Didactics: reason for questioning
	Didactics: use technology
	Didactics: asking why?
	Source of belief—didactic strategy
	Source of belief: understanding—university training
	Reason for caring: improve attitude
	Reason for caring: research
	Source of caring attitude: school
	Reflection: on errors
	Reflectivity: not
	Reflectivity in general
	Teaching style problems
Evidence and purpose of caring	Availability
	Conflict
	Encouragement
	Encouraging responses
	Personal care
	Positive dealing with wrong answer
	Re-explaining
	Understanding: faces

Martie

Martie believes that a good teacher's PMTI should be absolutely balanced in terms of subject specialisation, teaching-and-learning skills, and caring:

> I believe that in order to teach the best, you have to know your subject field. You have to be the best in what you do. In order to get your subject knowledge across to the learners it is important to be the best in your knowledge of the methods and processes used, especially in mathematics. I also believe that the level of learners' achievement are based mostly on their emotional, social and moral state. By being a positive influence here, you will increase their performance and achievement.

In the space of three sentences, she used the word "best" three times: "teach the best…be the best in what you do…be the best in your knowledge". She is confident of her subject knowledge: "I do consider myself, well, fairly knowledgeable on the subject of mathematics" and sets great store by her ability to answer every question correctly. As for teaching-and-learning skills, she says she acquired those while still a learner at school. She does acknowledge that she has since added to her repertoire in that regard. Her image of a good mathematics teacher she described as follows in the initial interview: "Well, it would have to be someone that's funny and obviously smart, um, but not too smart, someone that's able to convey what they're trying to teach, but effectively". When she was asked to describe her own teaching style, Martie immediately referred to the caring way she handled the learners: she saw the main thrust of her teaching as building up the learners while at the same time conveying knowledge. She believes in smiling often, being positive and encouraging.

At no point in the observed lessons does Martie falter or make a mistake in terms of the mathematical concepts that she is teaching. She explains with confidence and without hesitations, and explains repeatedly if she thinks that the learners do not quite understand. Her lessons are very structured: she requires her learners to write down "how to do" steps during the course of the lesson:

> Often I would tell them, "Write down in the little blocks somewhere in your books in colour pen or whatever. Make yourself a little note, here's like a little step for you 'how to do' the specific sum. So these are the steps you are going to use for most of them but you're going to modify them a little, but first you have to look for this then this and then this". And every time they asked me I would ask them, "Did you do the steps?"

Her teaching style is friendly and participatory: she continually invited learners to comment or provide information as to the topic she was teaching. She elicited choir-type responses by suffixing the expression "Né?" (Afrikaans equivalent for "Isn't that so?") after most statements she made in explaining a concept. In reacting to the learners' work as it was shown her, she nodded and smiled if it was correct, and shook her head, still smiling, if it was not. Not one of the learners who submitted incorrect answers appeared in any way crushed or defeated: on the contrary, all seemed eager to keep trying. However, the learners are at no point encouraged to discover anything on their own—she guides them very strongly at every step, saying things like "Nearly!", "Not quite!" or "Almost there!" in an effort to tweak learner responses into replicating her example.She is first and foremost a mathematics specialist; the videoed lessons show her as a caring, committed explainer (according to Ernest's (1988) model), who will go to any lengths to explain and re-explain until understanding, in her opinion, is evident:

> I did consider myself, well, fairly knowledgeable on the subject of mathematics. Obviously I've encountered quite a number of methods to teach … So at one section of the work, say now you're doing functions, I know how to teach about three or four different methods. So I have to do more than one method because I know that a lot of

> the children when I was in school didn't understand, necessarily like the one method, they understood more than one. So I like to use more than one but not…well obviously I try not to confuse the learners when I teach.

She certainly is reflective in that she tries to remediate didactical problems in a lesson and tries to plan for optimal coverage of the required material in the given time without sacrificing comprehension. She *cares*: but she has not yet mastered the technique of maintaining a professional distance without appearing to be more concerned about the topic than about the learner despite the fact that she says her main concern is to encourage, so that learners do not fall into the habit of feeling they are unable to succeed at mathematics:

> Because often children, especially at that age can feel that you break them down by saying "you're wrong!" because then they feel that "I'm stupid, I'm worth…well, if I'm stupid I'm worth nothing" and I'd rather not do that, I'd rather say "listen, you're worth a lot to me um, but try doing this differently, then you might get the right answer as well.

Her ease of use of the mathematics she teaches is evident in her classroom; her determination to ALWAYS be correct in what she teaches is clearly communicated both in what she says and in what she does in the classroom. It would seem that it is her aim and desire to put into practice the theory which makes sense to her: learner-centeredness and constructivist teaching strategies. She wants her learners to think and discover for themselves, but finds no way of leading them to the lesson outcomes without explaining and re-explaining while the learners remain passive, so they are *receivers* not *constructors* of the *correct* knowledge she shares with them. She sees herself as a Carer, but this aspect of her PMTI is not quite as significant in her teaching as Teaching-and-learning Specialist. Her priorities are clearly and hierarchically defined: she is a subject specialist first and foremost, she cares for the learners, and she is able to manipulate their thinking to comply with her version of what is right:

> I'm absolutely sure that I know what I'm doing. Um, I'm absolutely sure that I know that I have the best interest of every learner in front of me on my heart … Um, I know that and I know, well, I'm good at what I do. Um, I know what I'm teaching them is right um, and I know…I know I have people skills, I know how to work with them. *I know how to get the right responses that I want from them (emphasis added).*

She reflects on her practice, but seems currently unaware of the discrepancies between her teaching and her perception of herself as a constructivist facilitator of learning.

Ayesha

Ayesha sees herself primarily as a mathematics specialist, although she believes simultaneously and without apparent internal conflict, that the three given aspects of PMTI should rank *equally* in one's professional identity:

> I want to be a teacher who is able to focus on the subject knowledge and skills, the evaluation of the teaching and learning processes and moral development of learners. I feel that all these categories are equally important… I do see myself as caring but I feel that I want to be a subject specialist; I want to know my work. It's no use knowing how to teach when you don't know what you're teaching. So that's why I

> think that holds the most value, knowing your subject and then knowing how to teach.

She believes that didactical expertise is not something one can be taught: "…one can't really teach someone how to teach, I think it comes to you naturally". She is very critical of her own high school mathematics teacher who she described as traditional, teacher-centred, and uninvolved with the learners: "There used to be charts in her room but there was no learner-centeredness, there was no interaction—there was some kind of interaction but we used to be more scared of her than liking the subject". She describes herself as being passionate about mathematics and wanting to share that passion: "I want them to understand mathematics. I don't want them to just say "Step A, step B, and step C", which is just for them to get the marks. I give them the steps for them to get the marks but I also want them to understand mathematics and like it". Ayesha describes herself as someone who is also passionate about children. She explains that being a good mathematics teacher implies knowing one's learners and being able to take into account the diversity in the classroom as well as the personal difficulties which may be impediments to learning. She equated learner-centred teaching to being a caring educator, one who is also a moral preceptor and role model.

In the classroom, Ayesha is seen to teach procedure in a very structured way—she repeats steps and has the learners repeat steps as they watch her do sums on the board. She is able to teach the mathematical content confidently, and she certainly knows the procedures off by heart. While she is teaching, Ayesha does not move out of the space between the board and the teacher's desk which is almost against the board. She asks chorus-type questions such as "Do you all understand?" or "Side AB is equal to side BC, yes, no?" There is no other participation by the learners other than posing the odd question while they do exercises. She walks around the class checking that the learners are working, but she does not pause to interact with them in any way.

In terms of Ernest's (1988) model, Ayesha seems to be both an instructor and an explainer whose mission it is to transfer information as accurately and intelligibly as possible to her learners. Chorus-answer questions are used to assess learner comprehension. She also evaluates the expressions on their faces, looking for the Aha! moment of understanding. She knows her subject and seems comfortable answering the few questions that arise. However, questions outside of the "perimeter" of the lesson plan are not encouraged. She believes that a good lesson is an interactive one—yet her lessons are not designed to encourage learner participation. This apparent conflict is resolved when she explains that she makes a point of asking the class continually whether they understand. They answer in chorus. She describes this as interactive teaching. Ayesha is particularly concerned with discipline and believes that if she stops talking or allows the learners a freer participation, discipline will be lost. She describes a good teacher as a moral preceptor who cares about a learner's problems in as much as they inhibit learning: "Learners tend to look up to their teachers as role models. In order for the learners to be morally well developed, we as teachers need to be an ideal icon". She is friendly, without being particularly warm or caring in her attitude, being "strict but approachable".

Thandi

This is a prospective teacher who admits that her knowledge of mathematics in terms of what is required to be taught at high school is lacking. She compensates for this lack by reading up on what she is about to teach in various school textbooks. Although she states

that she loves mathematics and that she finds the subject difficult, she describes herself as a good mathematics teacher. She reasons that the fact that she has to learn the work alongside of her learners, makes her a better teacher. She prides herself on her knowledge of education theory that she became familiar with at university and the fact that she is always "up to date with everything". She says she developed her own classroom style without reference to either her tertiary training or her mentor teacher's guidance. She describes her teaching style as practical, based on the fact that she lets her learners work in class on the concept she has taught. Nevertheless, she tries to bring the learners to an enjoyment of the subject by leading them to link it to the real world and by trying to appear friendly. She states that being a caring teacher is "part of every teacher in each and every learning area, even mathematics. Learners are made up of their social space/world".

In the classroom, Thandi teaches while referring frequently to her file containing her lesson preparation. Her lesson is characterised by long uncomfortable pauses where she says nothing, makes no eye contact with the learners and enters into no interaction with the class at all. When asked what stands out for her in the viewing of the video, she said, "…the pauses…they take like…I take time…I take a lot of time". She does not react to inchoate murmurings of the class when something she has taught is not clearly understood. After just a few minutes, she stops and asks various learners to come and do a selection of sums on the board while she stands at the side of the classroom. When learners query what their classmate is writing, she says, "Don't ask me, ask him".

At no point does she answer any learner's questions or correct any of the mistakes that are made on the board.

In terms of Ernest's (1988) categories, she may be described as an instructor. Perhaps the most outstanding feature of her PMTI and its actualisation is her total lack of rapport with her learners, despite her theoretical consciousness of the necessity of the pastoral role in a good teacher's classroom practice. There appears to be no conflict between what Thandi understands to be a good mathematics teacher and her own practice: she believes herself to be a good teacher despite lacking in every one of the three aspects of PMTI. She believes her teaching to be learner-centred because she allows learners to teach; she thinks her lessons are thoroughly planned and that she "know[s] the stuff" despite the long awkward pauses in her lessons while she consults her file; she describes herself as a good teacher because she does *not* know the mathematics involved; she declares herself to be available and approachable to her learners, yet she holds herself aloof from them. Asked whether she believes in building relationships with the learners she stated emphatically, "No, I don't. No, I don't".

Thabo

Thabo appears to have a real love and appreciation for mathematics as a subject; in fact, it seems almost as if he is in awe of its magnitude compared with his knowledge of it. It is this enjoyment, he says, that he wants to share with his learners: he wants them to see what he sees. His desire is that the learners understand and are able to participate fully in the lesson and he declares his willingness to explain in various ways until understanding is achieved:

> Ok, myself as a mathematics teacher, I can say I'm a caring person and I have the time to listen to learners and I'm also approachable. Whenever they come to me and they ask me questions and then I'll…even after school I'll make time for them so that I can help them with those problems. If I'm not able to help them at that time, I'll

> promise them that when I go home I'll find information about that and then when I meet them again I'll explain it to them.

In his perception of his own PMTI, Mathematics Specialist and Teaching-and-learning Specialist are ranked equally. He believes in thorough and careful lesson preparation: this gives structure to the lesson, and helps in classroom management—the learners can see that you know what you are doing and where the lesson is headed:

> Now if you go to the class and then you are unprepared, you haven't planned the lesson, you won't know where to start and then when to give the uh, and then when to give the…the…the activity, when to ask the questions and the stuff. And then sometimes there are learners in the classroom that would always like to challenge you to see whether uh, to see whether you…you…you are knowledgeable in the subject. So, and if they can realise that you are not…you are not so knowledgeable on the top…on that topic, so they will start disrespecting you. So that helps in the classroom management also.

Thabo admits freely that he lacks the skills to counsel learners in a pastoral sense. He is however available for mathematical assistance at all times, even beyond school hours. He believes that his own learning process has just begun: he thought that, having taught mathematics while still at school that he had nothing further to learn about teaching—by his own admission he will not make that mistake again.

Thabo's lesson presentation is smooth, considered and well-prepared. He teaches without hesitation or error and answers learner questions with enthusiasm and confidence: "It is important to know or to have subject knowledge in order to be able to deliver it to your learners. It will be easy for me to teach if I have a sound knowledge of the subject" (sic). He explains extensively, using complex examples that appear to be beyond the learners' level of understanding. For example, he explained about hyperbolae being used to make lenses, and how these graphs could be used in lens design, analysis of capillary forces and rainbows, and the location of ships at sea prior to the use of global positioning systems. The learners showed interest in his explanations, but it was clear from the blankly puzzled looks on many faces that the idea of ship location was beyond their understanding. He frequently invites their participation by asking questions to be answered by individuals. In terms of Ernest's (1988) categories, Thabo is recognisable as both an explainer and a facilitator. Therefore, he explains as much as seems necessary and encourages the learners to ask questions and to express their understanding or lack thereof of the concepts he is teaching. However, his relationship in the classroom is with the subject first, and then with the learners, although he does not distance himself from them in any way. In his enthusiasm and passion for the subject, Thabo tends to pitch some of his teaching above the heads of his learners without noticing that he is doing so. He does however look at the body language of the learners to gauge whether they understand or not; he also asks questions eliciting individual participation—but he often answers the question himself without allowing the learners enough time to think. Upon watching the video, he recognised this himself:

> No, it…it…it was not enough time and uh, I've realised that in most cases uh, that is my problem because I don't give learners enough…enough time to think about what I've asked them. So I should work on that so…so that I can give them more time to think about it before I can explain that.

This he ascribes to the tight time schedule of the lessons.

John

Despite the fact that John ranks Mathematics Specialist as first in his PMTI, almost every statement John makes reveals the value he attaches to relationship as a springboard for effective teaching: relationships easily established on the sports field make teaching in the mathematics classroom easier:

> Ok, myself as a mathematics teacher. Well, obviously you need to know your content to know where you're going with your content, but just as equally I need to know how much I can push myself and how much I can interact with other kids and things like that. Being a pastoral role is for me the main thing (sic).

He believes in being creative and innovative in order to take the boredom out of mathematics lessons and to make them relevant to the learner's lived worlds:

> Let me explain it to you this way, "Take out your text books, turn to page this. Do this exercise". It sounds so boring, whereas if you've got something on the screen, now you say "Visualise this; explain to me how this happens. Look at this picture, what- if you rotate it this much, what happened to the picture, look how the dimensions change". There already you've just created a whole new perspective of mathematics and a whole new situation that…that can be derived from creativity. So, creativity is essential, for me, in mathematics, not necessarily easy but it needs to try and work its way in.

He believes that, if at all possible, all topics need to be linked to the real world so that learners can understand the usefulness of the knowledge and techniques they are acquiring through study of the subject. This linking is not always easy—for some topics, he says, "I'm still struggling to see the connection, but there is some sort of connection—you just need to always find it". He describes his style as interactive and spiced with humour. He also believes that teaching rules without reasons is futile, and that explanations he gives, need to answer the question, "why?" He believes in lesson planning, but only because it allows him to be flexible. He says it is impossible to plan a rigid structure for a lesson, because the dynamism of the class may change things.

In the classroom, John teaches with confidence and answers every question without hesitation or consultation with his file. At all times during his lessons, he has the learners "eating out of his hand"—their participation is keen and enthusiastic, yet discipline is not a problem. He uses humour to engage all the learners and real-world examples to elucidate the concepts he is explaining. The impartation of content knowledge during the lesson is done almost imperceptibly as the learners are guided into constructing it for themselves. His method of teaching largely entails posing leading questions, guiding the learners to a discovery of the knowledge about which the lesson revolves. At no time was he seen to be "lecturing" the class. In one of the observed lessons, he spent nearly a quarter of the lesson time drawing information from the learners about fractions. He poses both individual and chorus-answer questions.

He seems to be a facilitator, in terms of Ernest's (1988) categories. The most prominent aspect of John's PMTI is that of Carer. His classroom practice is based on relationship with his learners and the concern he has for them to do the very best they can under his aegis. Having been an introvert himself at school, he makes a point of getting to know the learners' names and drawing them out of their shells within the safety of his classroom. His intention is the establishment of mutual respect between himself and the learners, but he is aware of the dangers of over-familiarity:

> People respond more to their names and their personal being than anything "Hey you, boy", "Hey kid" or "Yes Sir" or something like that. So, like for instance, my second year prac, as soon as I got one of the boys' names he immediately opened up to me and then we started a whole learning thing going back and forth on the sports field. It's a very dangerous place to be as well as keeping the professional boundary, but it's a necessary place that you should be at.

His care for his learners goes beyond the exigencies of the classroom and the subject itself—he says he is there to help the learners with life in general as well.

Sipho

Sipho believes that being a subject specialist carries the greatest significance in his PMTI and is confident in his usage of mathematical concepts: he declares that he is "covered in that department". He sets great store by his knowledge of educational psychology which allows him to understand what the learners think and feel. He also believes that learners have a negative attitude towards mathematics that needs to be addressed. In describing a good mathematics teacher, he said that:

> A good mathematics teacher would be a teacher that's very professional and understands the subject, understands the psychology…the mentality that goes with mathematics. In order for you to teach it you have to understand it, you have to understand the whole idea, the feeling people have with this subject.

He also wants his classes to be fun, a strategy he says makes learners *want* to be attentive in his class:

> I can show them that this is going to be fun, just takes them away from "Ah, mathematics is so boring! Numbers again", Make it a little social; make them see it in a social way, more like chatting, talking to me and asking me about mathematics, the actual content, reality.

He is concerned about shy or reticent learners. He points out that it is useful to allow the learners to teach from time to time: for this task he selects learners who seek attention or approval: "Let him explain rather than me talking, see that he knows something and he wants to share something, let him explain. Let's see how he does it …then I incorporate it to the others so that I can accommodate everyone (sic)". In this way, he says, he is able to satisfy the need of the learner, while at the same time finding out what such a learner knows. He is driven to facilitate relationships and cultural respect, having himself been the victim of a mathematics teacher's racist comments at high school.

In the classroom, Sipho is seen to teach the topic at hand with ease and comfort, but not without errors. He uses any object to hand to illustrate a concept, as well as teaching strategies that he describes as "dramatic", and that include walking up and down the class making large gestures with his hands: "a different audience and you present it differently. It…it all boils down to…to the…to the audience, how they respond". Sometimes class discipline is sacrificed in his application of these strategies. He plans his lessons in order to be organised, but responds readily to questions that may seem less closely related to the topic at hand. He also tries to make the work relevant to the everyday lives of the learners: for example, in teaching about ratio, he used the demographics of the classroom to illustrate comparisons.

In terms of Ernest's (1988) categories, he is a facilitator. He is driven to facilitate relationships and cultural respect:

> You know, just because you're a black teacher and I'm still young they didn't really take me seriously so I had to go the extra mile to prove it to them that I can teach this and I have the skill to help you to understand.

He wants to please, and to be liked and accepted by his learners. He describes himself as someone who knows and loves his subject and who knows and loves the learners, particularly in view of their cultural diversity, so his teaching-and-learning role is characterised by his determination to integrate these two loves—the subject and the learners. It would seem that his primary strategy is to make the learners enjoy being with him in the class. Part of this strategy implies a negotiation of meaning in the actual content of the lesson: he strives to draw information from the learners by asking questions and prompting them to access the prior knowledge they might have to be able to do the work at hand. It is evident in what Sipho does in the classroom that he is primarily a carer, not a subject specialist.

Discussion

Often the espoused theory and the theory-in-use observed in the study of the six participants were not congruent. It may be concluded that, if Palmer (2007) is correct and actually "we teach who we are" (p. 2), then the "who we are" of four out of these six prospective teachers is not who they say they are, judging by their teaching.

Mathematics expertise

These six prospective teachers present varying levels of mathematics expertise. Thandi, for example, finds the work difficult and has to research the content very carefully before each lesson. She also does not seem to perceive mathematical errors in the work done on the board by learners. Thabo, Ayesha, Martie, John, and Sipho express confidence and demonstrate ease in handling both the mathematical content and the questions that arise in the classroom. When appraising their mathematical expertise in terms of their ease of use of mathematical concepts, it would seem that all six have a fairly accurate idea of themselves in this area: what they say they are seems to be very close to what can be observed in the classroom. We suggest that the match between their perception of themselves as Mathematics Specialists and their manifested mathematical expertise is ascribable to the clear communication of mathematical competence through marks given which indicate performance in mathematics assessments: the prospective teachers know exactly well how they fared in the mathematics modules at university.

Teaching-and-learning strategies

In this aspect of PMTI, there was less congruency between espoused theory and theory-in-use. John, like Sipho, leads the learners to understanding of the concept he is teaching; the other four participants have a formal approach to the teaching of mathematics, tending not to involve the learners in the actual 'teaching' part of the lesson. It is therefore not surprising that these four prospective teachers also tend to be Explainers. All six have adopted strategies that they believe facilitate learning: Martie explains and re-explains,

recommending that the learners take notes; Ayesha believes in drilling concepts; Thandi tries to relate concepts to the real world, and allows learners to explain their understanding by doing examples on the board; Thabo also tries to link mathematics to the real world, asking questions, but without giving the learners enough time to think and answer before he answers his own question; John starts a lesson with an "attention-grabber" and proceeds to teach using leading questions, the answers of which yield facts to be written in the "knowledge box" area on the board; Sipho walks around the class, talking, waving his arms about, stopping to chat to individuals while he checks their work, all to make his lessons fun and to keep the learners attentive. All the participants were, in varying degrees, aware of the diversity that categorises South African classrooms. However, other than explaining and re-explaining, the strategies of which these prospective teachers speak for dealing with diversity are largely not visible. Sipho, however, implements different languages in his explanations in the hope of improving understanding.

Evidence of understanding

These six participants have also come to accept certain indicators as evidence that the learners have understood the concepts in question. Ayesha, Martie, Thabo, and John take note of the expression in the learners' eyes. Martie listens to the kind of questions the learners pose, while Ayesha, Thabo, John, and Sipho pay particular attention to learner responses to chorus-answer questions *they* pose. Martie finds homework useful as an indicator of understanding; Thabo does not—he finds the learners do not do the homework, so he has to help them finish the work in class anyway. Martie, Ayesha, and Thandi do "classroom patrol", checking learners' books as they work. However, in all six cases their lack of experience has made them unable to determine whether these techniques give them an accurate picture of the actual levels of learner understanding in their classes. In fact, in four out of the six cases what was claimed to be evidence of understanding was observed to be inaccurate: the learners were observed to be making mistakes (Martie's class), trying to help each other rather than ask for help (Ayesha's class), asking for help but being ignored (Thandi's class) and not being given enough time to answer questions (Thabo's class).

Teacher/learner-centeredness

All six participants espouse learner-centeredness as a theory. However, they do not seem to have the same understanding of what learner-centeredness means. To Ayesha and Martie, it means the learners answer questions she poses; Thandi believes that her classroom is learner-centred because learners do sums and explain their work on the board. Thabo experiences difficulty in making his lessons learner-centred, which he believes would be the right thing to do, but he does not think he has mastered the skills to do so yet. John and Sipho involve learners from the beginning to the end of the lesson. The other four participants give little opportunity for learner discovery: they teach and explain, answer what questions there are and give exercises to be done as classwork. Two possible reasons are postulated for this disparity between belief and practice: these prospective teachers may believe that they are in fact operating in a learner-centred way; or what they *say* they believe, and how they believe they should *act* are not integrated effectively in their PMTI's.

Flexibility/rigidity

In planning their lessons, Martie, Sipho, John, and Thabo believe that they are providing a structure from which they can deviate if necessary. Ayesha uses her lesson planning to make sure the learners are busy all the time—flexibility is not an option. Thandi does not plan her lessons to be flexible. Her planning is rigid around the content that she has prepared for the lesson because she is not confident of mathematical content beyond her preparation. Much the same picture presents itself when it comes to evidence and purpose of caring.

Evidence and purpose of caring

This aspect of PMTI shows the least congruence between espoused theory and theory-in-use. Thandi, for example, advocates the theory of the nurturing role to be played by a teacher, but she declares that she has no intention of implementing any such role if it requires more time than is available in class. Thabo and Ayesha are concerned about the learners for the sake of the mathematics on the grounds of learners not being able to learn if they are upset, but there is no evidence in their classrooms of care of the learners for their own sake. Martie and John do however appear to care for the learners for their own sake, while Sipho is driven to make sure all learners realise that he values them equally whatever their race.

These prospective teachers thus demonstrate that while they may certainly be teaching who they are, this is not necessarily who they *think* they are. They may believe that they are Mathematics Specialists, Teaching-and-learning Specialists, and Carers, but when they are observed at work in the classroom these specialisations are not necessarily, or at least not consistently evident.

Conclusions

These prospective teachers' perceptions of their own PMTI and the manifestation of that PMTI in the classroom are not consistently congruent. Using the terminology of Argyris and Schön (1974), their espoused theory and theory-in-use are not the same. If we accept that "we teach who we are", it must be with the understanding that this is not necessarily "who we say or think we are". For example, Ayesha is critical of her teacher-centred high school teacher and speaks highly of the theory of learner-centred teaching in the mathematics classroom, but her own classroom practice is entirely teacher-centred; Thandi espouses a holistic understanding of learners who, she says, should be seen in terms of their whole world, yet she involves herself in no way with any part of the learners' world.

We propose two reasons for this. It may be that these prospective teachers are not yet capable of true reflection on themselves and their own practice. This implies that Beijaard et al.'s (2000) assumption that teachers' *perceptions* of their PMTI reflects *knowledge* of their PMTI does not apply to these prosepective teachers. It may also be that PMTI is contextually bound—that the "saying" occurs in a different context to the "doing" when it comes to describing one's PMTI and actualising it in the classroom. We therefore suggest that the definition of PMTI "who I am at this moment in this context" is one that encompasses the possibility of incongruence as a result of different contexts.

If this incongruence between espoused theory and theory-in-practice is to be addressed, "reflective practice" needs to be more than another theory that is taught: prospective

teachers will need to be guided into practical means of reflection on their own practice. This study proves that incongruence between *perceptions* of PMTI and *actual* PMTI as manifested in classroom practice is a real possibility, in which case *research of professional teacher identity which does not include observation of the teacher-in-action is incomplete*. Researchers in this area need to be conscious of the fact that what is described in personal narratives regarding PMTI may be idealistic rather than real, unless the individual is a truly reflective practitioner.

This suggests that research into professional teacher identity which is reliant only on the individual's perceptions of their own identity is, in fact, only looking at half the picture. It is essential to observe those individuals in the classroom—it is very possible that who they say they are as teachers is not in fact who they actually are in the classroom. The definition of PMTI as given in this study, "who I am at this moment in this context" points to the significance of context: the saying and the doing of "who I am" take place in two different contexts, which is directly linked to what is espoused and what is put into action.

References

Ainley, J., & Luntley, M. (2007). The role of attention in expert classroom practice. *Journal of Mathematics Teacher Education, 10*, 3–22.

Arends, F., & Phurutse, M. (2009). *Beginner teachers in South Africa: School readiness, knowledge and skills*. Cape Town: HSRC Press.

Argyris, C., & Schön, D. (1974). *Theory in practice: Increasing professional effectiveness*. London: Jossey-Bass.

Ball, D. L., Thames, M. H., & Phelps, G. (2008). Content knowledge for teaching—What makes it special? *Journal for Teacher Education, 59*(5), 389–407.

Beauchamp, C., & Thomas, L. (2009). Understanding teacher identity: An overview of issues in the literature and implications for teacher education. *Cambridge journal of Education, 39*(2), 175–189.

Beijaard, D. (2010). Personal communication.

Beijaard, D., Meijer, P. C., & Verloop, N. (2004). Reconsidering research on teachers' professional identity. *Teaching and Teacher Education, 20*, 107–128.

Beijaard, D., Verloop, N., & Vermunt, J. D. (2000). Teachers' perceptions of professional identity: An exploratory study from a personal knowledge perspective. *Teacher and Teacher Education, 16*(2000), 749–764.

Bohl, J. V., & Van Zoest, L. R. (2002). Learning through identity: A new unit of analysis for studying teacher development. In A. D. Cockburn & E. Nardi (Eds.), *Proceedings of the international group for the psychology of mathematics education annual meeting* (Vol. 2, pp. 137–144). Norwich: University of East Anglia.

Bromme, R. (1991). Wissenstypen und professionnelles Selbstverstandniss. *Zeitschrift fur Pädagogik, 37*, 769–785.

Cardelle-Elawar, M., Irwin, L., & de Acedo Lizarraga, M. L. S. (2010). Looking at teacher identity through self-regulation. *Psicothema, 22*(2), 293–298.

Chevrier, J., Gohier, C., Anadon, M., & Godbout, S. (2007). Construction de l'identité professionnelle des futures enseignantes: Dispositifs de formation presents et souhaités selon les acteurs responsables de la formation des maîtres au préscolaires et au primaire. In C. Gohier (Ed.), *Identités prefessionnelles d'acteurs de l'enseignement*. Québec: Presses de l'Université de Quebec.

Chin, C. (2006). Classroom interaction in science: Teacher questioning and feedback to students' responses. *International Journal of Science Education, 28*(11), 1315–1346.

Cohen, L., Manion, L., & Morrison, K. (2000). *Research methods in education*. London: Routledge Falmer.

Coldron, J., & Smith, R. (1999). Active location in teachers' construction of their professional identities. *Journal of Curriculum Studies, 31*(6), 711–726.

Creswell, J. W., & Miller, D. L. (2000). Determining validity in qualitative enquiry. *Theory into Practice, 39*(3), 124–130.

Darling-Hammond, L., & Richardson, N. (2009). Research review/teacher learning: What matters? *How Teachers Learn, 66*(5), 46–53.

Day, C. (2002). School reform and transitions in teacher professionalism and identity. *International Journal of Educational Research, 37*, 677–692.

Department of Education. (2006). *The national policy framework for teacher education and development in South Africa*. Retrieved February 19, 2011, from www.info.gov.za/view/DownloadFileAction?id=70084.

Ernest, P. (1988). The impact of beliefs on the teaching of mathematics. In *Paper presented at the ICME IV*, Budapest, Hungary.

Fisher, D., & Frey, N. (2007). *Checking for understanding—Formative assessment techniques for your classroom*. Alexandria: ASCD.

Flores, M. A., & Day, C. (2006). Contexts which shape and reshape new teachers' identities: A multi-perspective study. *Teaching and Teacher Education, 22*, 219–232.

Gee, J. P. (2000). Identity as an analytic lens for research in education. *Review of Research in Education, 25*, 99–125.

Gibbs, G. (2007). *Analyzing qualitative data*. London: Sage.

Graven, M. (2004). Investigating mathematics teacher learning within an in-service community of practice: The centrality of confidence. *Educational Studies in Mathematics, 57*, 177–211.

Hamachek, D. (1999). Effective teachers: What they do, how they do it, and the importance of self-knowledge. In R. P. Lipka & T. M. Brinthaupt (Eds.), *The role of self in teacher development* (pp. 189–224). Albany, NY: State University of New York Press.

Hill, H. C., Rowan, B., & Ball, D. H. (2005). Effects of teachers' mathematical knowledge for teaching on student achievement. *American Educational Research Journal, 42*(2), 371–406.

Hodgen, J., & Askew, M. (2007). Emotion, identity and teacher learning: Becoming a primary mathematics teacher. *Oxford Review of Education, 33*(4), 469–487.

Korthagen, F. A. J. (2004). In search of the essence of a good teacher: Towards a more holistic approach in teacher education. *Teaching and Teacher Education, 20*, 77–97.

Kunter, M., & Baumert, J. (2006). Who is the expert? Construct and criteria validity of student and teacher ratings of instruction. *Learning Environments Research, 9*, 231–251.

Lave, J., & Wenger, E. (1991). *Situated learning: Legitimate peripheral participation*. Cambridge, England: Cambridge University Press.

Leatham, K. R., & Hill, D. S. (2010). Exploring our math complex identities. *Mathematics Teaching in the Middle School, 16*(4), 224–231.

Maxwell, J. A. (1996). *Qualitative research design*. Thousand Oaks, CA: Sage.

National Council of Teachers of Mathematics. (2008). *Response to national math panel: The importance of effective teachers*. Author. Retrieved July 22, 2011, from http://www.nctm.org/news/content.aspx?id=14391.

Palmer, P. J. (2007). *The courage to teach: Exploring the inner landscape of a teacher's life*. San Francisco: Jossey-Bass.

Pang, J. (2009). Good mathematics instruction in South Korea. *Zentralblatt für Didaktik der Mathematik, 41*, 349–362.

Shapiro, S. (2010). Revisiting the teachers' lounge: Reflections on emotional experience and teacher identity. *Teaching and Teacher Education, 26*, 616–621.

So, W. W. M., & Watkins, D. A. (2005). From beginning teacher education to professional teaching: A study of the thinking of Hong Kong primary science teachers. *Teacher and Teacher Education, 21*, 525–541.

Stronach, I., Corbin, B., McNamara, O., Stark, S., & Warne, T. (2002). Towards an uncertain politics if professionalism: Teacher and nurse identities in flux. *Journal of Educational Policy, 17*(1), 109–138.

Thompson, A. G. (2009). The relationship of teacher's conceptions of mathematics teaching to instructional practice. In A. J. Bishop (Ed.), *Mathematics education—Major themes in education* (pp. 58–76). New York: Routledge.

Van den Berg, R. (2002). Teachers' meanings regarding educational practice. *Review of Educational Research, 72*(4), 577–625.

Van Es, E. A., & Sherin, M. G. (2008). Mathematics teachers' "learning to notice" in the context of a video club. *Teaching and Teacher Education, 24*, 244–276.

Van Zoest, L. R., & Bohl, J. V. (2005). Mathematics teacher identity: A framework for understanding secondary school mathematics teachers' learning through practice. *Teacher Development, 9*(3), 315–345.

Varghese, M., Morgan, B., Johnston, B., & Johnson, K. A. (2005). Theorizing language teacher identity: Three perspectives and beyond. *Journal of Language, Identity and Education, 4*(1), 21–44.

Vloet, K., & van Swet, J. (2010). I can only learn in dialogue! Exploring professional identities in teacher education. *Professional Development in Education, 36*(1), 149–168.

Volkmann, M. J., & Anderson, M. A. (1998). Creating professional identity: Dilemmas and metaphors of a first-year chemistry teacher. *Science Education, 82*(3), 293–310.
Walkington, J. (2005). Becoming a teacher: Encouraging development of teacher identity through reflective practice. *Asia-Pacific Journal of Teacher Education, 33*(1), 53–64.
Wenger, E. (2000). Communities of practice and social learning systems. *Organization, 7*(2), 225–246.
Wilkins, J. L. M. (2008). The relationship among elementary teachers' content knowledge, attitudes, beliefs, and practices. *Journal of Mathematics Teacher Education, 11*, 139–164.
Zembylas, M. (2003). Interrogating "teacher identity": Emotion, resistance and self-formation. *Educational Theory, 53*(1), 107–127.